# Getting Leverage on Normativity

Anson Berns

Draft 8/26

# Motivation: Learning Normativity

**Learning Normativity desiderata** Demski 2020

- Whole-process, all-levels learning
- The meaning of any feedback can later be reinterpreted
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**Goal:** build a solution to these desiderata using logical induction as a key ingredient

# The law

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**A paradigm domain for Learning Normativity**

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It's not enough to predict human judges, we want to learn the underlying norm.

How should AI put a credence on **"John Smith should be found guilty"**?

# A middle way

Even though there's **no "final answer"**, our judgments are **not arbitrary**.

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**Leverage**: how are normative beliefs **constrained**, without being pinned to a single number?

# Two kinds of leverage

**Coherence**

How do normative claims bear on each other?

**Groundedness**

How do empirical claims bear on normative claims?

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- Similar cases get similar verdicts
- Doctrine and precedent
- Internal logic of the law

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Particular coherence and groundedness claims are themselves up for debate

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- The only kind of external **feedback** available to a vanilla logical inductor is immutable certainty from deduction
- We are lacking a representation of **reasons**, the network of non-logical relationships between sentences

# What logical induction is missing

As a solution to the Learning Normativity desiderata, logical induction is missing an account of **leverage**.

# An afoundationalist picture

**Foundationalist accounts**

- **Metaphor:** A layer of immutable bedrock underpinning every story of a tall building
- Based on **self-justifying certainty**, deepest principles that can't be questioned

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# An afoundationalist picture

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- Based on **revisable pre-licensed entitlements**, an initial induction into the normative domain that can be questioned after learning
- Defeasible bridge principles that let us go beyond coherentism

# A bounded reasoner

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**Credence states**

$$\mu \in \Delta(\Omega_t) \qquad\qquad \mu(\varphi) = \sum_{\substack{\omega \in \Omega_t \\ \text{with } \omega \models \varphi}} \mu(\omega)$$

# Warrants

- **New primitive:** a **warrant** $\rho : \ \alpha(n) \rightsquigarrow \beta(n)$
  - "$\alpha$ sentences provide support for $\beta$ sentences because of defeasible reason $\rho$"
  - One schema that applies to multiple cases, indexed by $n$

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- Warrants themselves are strengthless, numbers are going to enter contextually (soon)

# Undercutting and rebuttal

This Horty-inspired setup distinguishes two kinds of **conflict** between reasons

**Rebuttal**

$$\rho : \ \alpha(n) \rightsquigarrow \beta(n)$$

$$\rho' : \ \gamma(n) \rightsquigarrow \neg\beta(n)$$

**Undercutting**

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$\alpha$: "Alice says she saw John do it"

$\beta$: "John should be found guilty"

$\gamma$: "Bob says John was with him"

### Undercutting

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$\alpha$: "Alice says she saw John do it"

$\beta$: "John should be found guilty"

$\delta$: "Alice was drunk"

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**Endorsed gambles**

Desirable gambles: Walley 1991

Equivalently: endorsing $E$ = accepting a **one-sided desirable gamble**

$$\mu(\varphi \wedge \psi) \geq c \cdot \mu(\varphi) \quad \Longleftrightarrow \quad \mu(g_E) \geq 0, \;\; g_E = \mathbf{1}_{\varphi \wedge \psi} - c \cdot \mathbf{1}_{\varphi}$$

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# Three ways to endorse a warrant

Fix a $\rho$ and $n$ and abbreviate: $X = \alpha(n) \qquad A = \mathrm{App}_\rho(n) \qquad Y = \beta(n)$

**Three kinds of endorsements** conditional form

| | | |
|---|---|---|
| $(E_P)$ | premise | $\mu(X) \geq c_P$ |
| $(E_A)$ | applicability | $\mu(A \mid X) \geq c_A$ |
| $(E_S)$ | strength | $\mu(Y \mid X \wedge A) \geq c_S$ |

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division-free form

$(E_P)$ premise $\quad \mu(X) \geq c_P$

$(E_A)$ applicability $\quad \mu(X \wedge A) \geq c_A \cdot \mu(X)$

$(E_S)$ strength $\quad \mu(X \wedge A \wedge Y) \geq c_S \cdot \mu(X \wedge A)$

# Leverage propagation

For every compatible credence state $\mu$:

$$\mu(Y) \;\geq\; \mu(X \wedge A \wedge Y) \quad \text{(monotonicity)}$$

Suppose for some $\rho$ and $n$ we have all three endorsements:

$$\mu(X) \geq c_P$$

$$\mu(X \wedge A) \geq c_A\, \mu(X)$$

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Endorsements of warrants multiplicatively propagate numerical bounds from one sentence to another

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"Every compatible credence state $\mu$" means every $\mu \in \mathcal{C}_t$, distributions inside the **credal region** generated by the current endorsement book $\mathcal{E}_t$

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**Three levels of demandingness**

| | | |
|---|---|---|
| $\mathcal{C}_t = \Delta(\Omega_t)$ <br> the book forces nothing | $\emptyset \neq \mathcal{C}_t \subsetneq \Delta(\Omega_t)$ <br> constrained without being pinned | $\mathcal{C}_t = \emptyset$ <br> the book demands irrationality <br> "sure loss" |

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As the intersection of finitely many halfspaces, $\mathcal{C}_t$ is a polytope, which lets us use the math of **linear programming**

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Each endpoint is the value of a finite linear program over $\mathcal{C}_t$, so the min and max are attained

# Worked example

At John's murder trial, he's accused of being seen in a bloody shirt:
$\text{Bloody}(\mathrm{John}) \rightsquigarrow \text{Guilty}(\mathrm{John})$

- ($c_P$) "Was John actually bloody?"
- ($c_A$) "Given that he was bloody, is that fact relevant to his guilt?"
- ($c_S$) "Given that he was bloody without extenuating circumstances, how strong a reason is that for his guilt?"

Take $(c_P, c_A, c_S) = (0.9,\ 0.9,\ 0.9)$:

$$\mu(\text{Guilty}(\mathrm{John})) \in [0, 1] \xRightarrow{\text{endorsements}} \mu(\text{Guilty}(\mathrm{John})) \in [\,0.729,\ 1\,]$$

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## Remarks

- LP says $\mu = 0.729$ is attained
- A warrant like $X \rightsquigarrow \neg Y$ can give us an upper bound below 1

# LP duality

Linear programming provides a **dual certificate** $(\lambda, h)$ for $\underline{P}_t(\varphi)$:

$$\mathbf{1}_\varphi - c \; = \; h + \sum_{E \in \mathcal{E}_t} \lambda_E \, g_E, \qquad \lambda_E \geq 0, \;\; h \geq 0$$

$c$: the floor being certified  $g_E$: accepted gambles  $\lambda_E$: weights  $h$: slack

"In every world, $\varphi$ pays out exactly enough to cover $c$ in cash plus a bundle of bets the book already accepts, with $h$ left over."

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Apply $\mu \in \mathcal{C}_t$ to both sides:

$$\mu(\varphi) - c \;=\; \mu(h) + \sum_{E \in \mathcal{E}_t} \lambda_E\, \mu(g_E) \;\geq\; 0$$

A certificate demonstrates $\mu(\varphi) \geq c$ for every compatible $\mu$

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### A place for whole-process feedback

The $E$ with $\lambda_E > 0$ are the reasons cited; the dual solution is the written opinion

$\lambda_E$ lets us receive feedback about mixtures of lots of different reasons

# LP duality

**Off-the-shelf machinery** — Lukasiewicz 2001

- Our division-free bounds are **conditional constraints** $\mu(\psi \land \varphi) \geq \ell \cdot \mu(\varphi)$
  - An endorsement compiles to an already-understood object
- Two LPs compute $\underline{P}_t,\ \overline{P}_t \qquad \mathcal{C}_t \neq \emptyset \iff$ feasibility (Farkas' Lemma)
  - The natural problems come pre-solved

# Statics recap

$$\underbrace{\rho:\ \alpha(n) \rightsquigarrow \beta(n)}_{\text{relevance}} \quad \text{and} \quad \underbrace{\mathcal{E}_t}_{\text{strength}} \longrightarrow \underbrace{\mathcal{C}_t}_{\text{geometry}} \longrightarrow \underbrace{[\,\underline{P}_t(\varphi),\ \overline{P}_t(\varphi)\,]}_{\text{leverage}} \quad \text{with} \quad \underbrace{(\lambda, h)}_{\text{justification}}$$

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**Epistemic graphs** Hunter–Polberg–Thimm 2020

- Linear constraints on credences in probabilistic argumentation
- Change over time only as one-shot belief revision: no settlement, no learning

# Statics recap

Can we now move towards a **dynamics** of leverage?

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Two warrants $\rho : \alpha(n) \rightsquigarrow \beta(n)$ and $\sigma : \gamma(n) \rightsquigarrow \beta(n)$, each endorsed at the same $(c_P, c_A, c_S)$:

$$\mu(\beta(n)) \geq c_P\, c_A\, c_S \quad (\text{via } \rho) \qquad\qquad \mu(\beta(n)) \geq c_P\, c_A\, c_S \quad (\text{via } \sigma)$$

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### Conflict

Undercutter: $\delta(n) \rightsquigarrow \neg\mathrm{App}_\rho(n)$

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## Accrual

Should $\rho$ and $\sigma$ together mean a tighter bound than either alone?

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**Warrants with the same static geometry can have different dynamics**

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A sharpened **desideratum**: any LI-based solution to Learning Normativity must have prices that respond to **leverage**

# Warning

Claims from this point on rest on LLM-assisted work

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- Make the trader rich? **The market can't tell wealth from evidence**

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- Declare the sentence settled? **Goes against the whole point**
- Add a trader representing the bound? **They'll go broke**
- Make the trader rich? **The market can't tell wealth from evidence**

**Where can force enter?** Three LI objects we could modify:

| Settlement | Traders | Market Maker |
|---|---|---|
| Abandons the phenomenon | Needs unlimited external funding | Most promising idea |

# Modifying the market maker

**New rule:** the market maker's quotes must respect the current book, $p_t \in \mathcal{C}_t$

Can traders exploit this modification?

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### Operative-Force Wealth Bound

$$\textbf{winnings} \leq \textbf{slack} + \textbf{rate} \times (\textbf{risk} + \textbf{movement})$$

$$G_N(w) \;\leq\; \varepsilon_{1:N}/\theta \;+\; \big((1-\theta)/\theta\big)\,\big(R + M_N\big)$$

If the book always leaves the market maker at least a $\theta$-share of its freedom to quote, then it's not exploitable.

# Answerability

Now endorsements constrain prices, what constrains endorsements?

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## Mechanism

Challengers post a bond and issue **challenges** against particular commitments of the book, governed by checkable rules. The book pays **charges** to the challenger each period until it **answers**: refuting the challenge under the rules, or revising through an authorized route. A book is **answerable** if every challenge eventually ends, every revision travels an authorized route, and the total charges ever paid are finite.

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**Diachronic criterion:** you must be answerable to your past self

# Learning without ground truth

What constitutes improvement when there is no objective standard?

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What constitutes improvement when there is no objective standard?

### $\Phi$-regret

Greenwald–Jafari 2003

How much loss $\ell_t$ can be reduced from action $a_t$ within a single edit class $\Phi$:

$$R_T^{\Phi} = \sup_{\varphi \in \Phi} \sum_{t \leq T} \left[ \ell_t(a_t) - \ell_t(\varphi(a_t)) \right]$$

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**$\Phi$-regret** Greenwald–Jafari 2003

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$$R_T^{\Phi} = \sup_{\varphi \in \Phi} \sum_{t \le T} \left[ \ell_t(a_t) - \ell_t(\varphi(a_t)) \right]$$

Take $\ell_t$ = charges owed and $\Phi$ = the **lawful** edits of the book

**Lawfulness:** part global condition (**answerability**) and part local condition (**reasons-responsiveness**)

# Status of LLM work

- **Force:** not exploitable while a core of quoting freedom is preserved
- **Answerability:** an explicit policy is answerable from any authorized start; minimum funding exact
- **Composition:** both pressures on one history; no interference as long as funding is kept separate
- **Identity:** obligations survive vocabulary change; nothing merges silently
- **Demand:** bounded liability forces rulings to track admitted arrivals
- **Settlement:** an interface world-reports must meet before they get force
- **Learning:** candidate criterion; staged proof plan

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## In the repo

Finite models in Python; hand proofs of theorems; one script checks every claim

# Open questions

- What precisely is "reasons-responsiveness"?
- What's the real relationship to LI? Is this a new LI-like construction, or an extension of LI?
- What should demand optimize for? The existing **docket** story just gives a feasible region
- Where exactly do endorsements come from? Candidate answer: **inquiry**

# Conclusion

The **concept** of leverage is what's missing from logical induction for normativity

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$\longrightarrow$

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# Conclusion

| | | | | |
|---|---|---|---|---|
| The **concept** of leverage is what's missing from logical induction for normativity | $\longrightarrow$ | The **statics** of leverage use known math for a concrete desideratum | $\longrightarrow$ | The **dynamics** of leverage give rise to hard, but tractable problems with open design questions |

# References


- A. Demski. *Learning Normativity: A Research Agenda.* Alignment Forum, 2020.
- A. Greenwald and A. Jafari. A general class of no-regret learning algorithms and game-theoretic equilibria. *COLT*, 2003.
- A. Hunter, S. Polberg, and M. Thimm. Epistemic graphs for representing and reasoning with positive and negative influences of arguments. *Artificial Intelligence* 281:103236, 2020.
- T. Lukasiewicz. Probabilistic logic programming with conditional constraints. *ACM Transactions on Computational Logic* 2(3):289–339, 2001.
- S. Nair and J. Horty. The logic of reasons. In D. Star (ed.), *The Oxford Handbook of Reasons and Normativity*. Oxford University Press, 2018.
- P. Walley. *Statistical Reasoning with Imprecise Probabilities.* Chapman & Hall, 1991.